

ICT/0Levels

English Medium

Learn ICT with Shawn sir.

Grade 10

5 *Logic Gates*



What are Logic Gates

Draw 3 Basic Logic Gates and 2 Combined Logic Gates

AND

OR

NOT

NAND

NOR

Draw the following Logic Gates using the Algebraic equation

$$(C \cdot D)$$

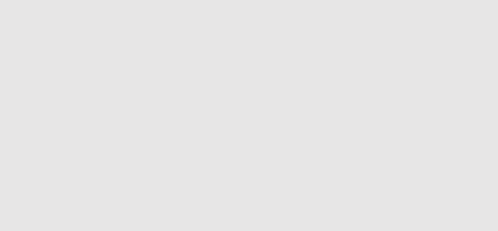
$$\overline{(A + B)} + (B \cdot C)$$

$$(A + C) + C$$

$$(A + B) + (A \cdot B)$$

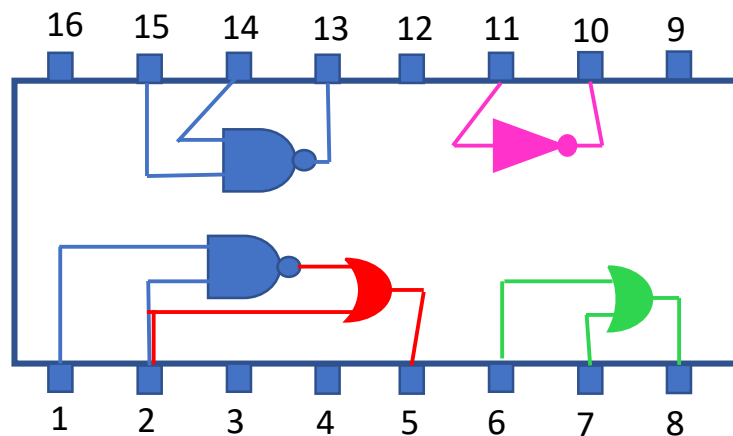


[illegible]



[illegible]

Find the output of the given pin when the correct input voltage is supplied to their respective pins.



-

Real-World Application

An excellent security system should activate an alarm only when:

- The front-door is open ($A = 1$)
- The back-door is open ($B = 1$)
- The window is open ($C = 1$)

Which logic gate should be used
(Draw the logic gate with the A B inputs)

Real-World Application

A technical student built an Arduino based temperature Fan Regulation system for his class. The Fan will activate only when:

- The Fan switch is ON ($A = 1$)
- The windows is opened ($B = 0$)
- The temperature is above 25°C ($C = 1$)

Which combination of logic gate should
be used (draw the logic gates with all inputs)

Real-World Application

School Computer Login System:

A student can access a computer only when:

- The correct Username is entered ($A = 1$)
- The correct Password is entered ($B = 1$)

Which logic gate best represents this situation?
Draw a TRUTH table for this.

Real-World Application

School Science Laboratory

A chemical cabinet should unlock only when:

- The teacher's key card is detected ($A = 1$)
- The fingerprint scan is successful ($B = 1$)
- Time is between 7AM and 9PM ($C = 1$)

However, if anyone is false, the cabinet must remain locked. Which combinational logic gates should control the lock?

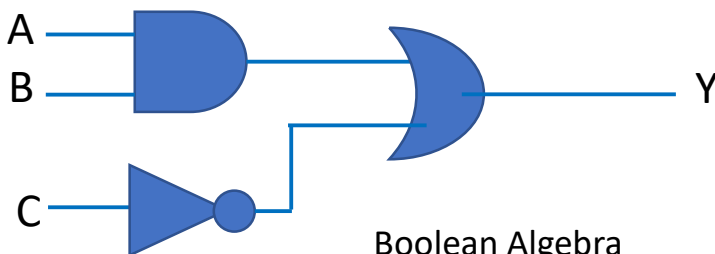


Logic Gates to algebraic equation

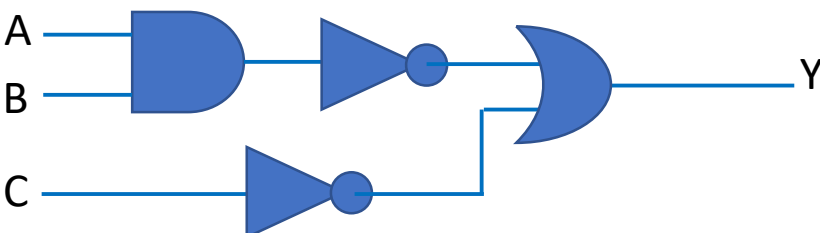
A mega-factory in Japan is constructing a robot assembly and requires the following Logic Gates to be documented in their computer system. As of now, the photo-lithography design system can only accept Boolean expressions. Convert the following Logic Gate diagrams to Boolean algebra.



Boolean Algebra _____



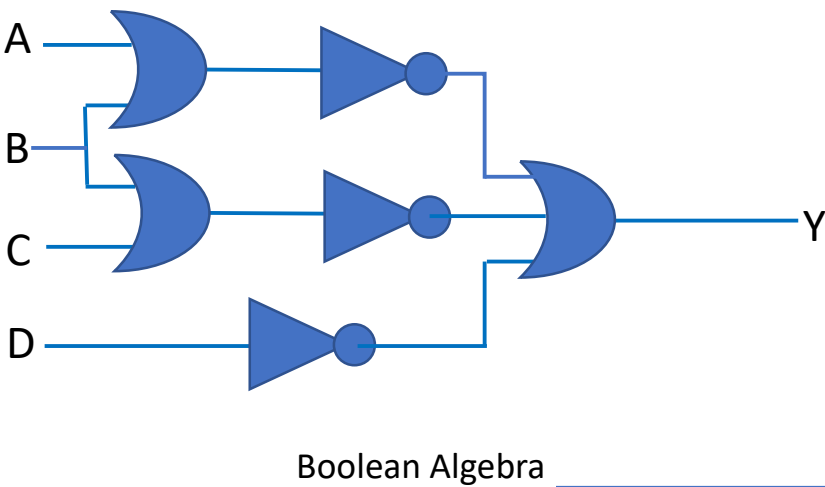
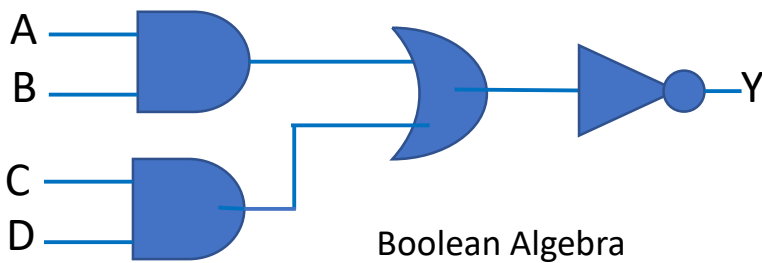
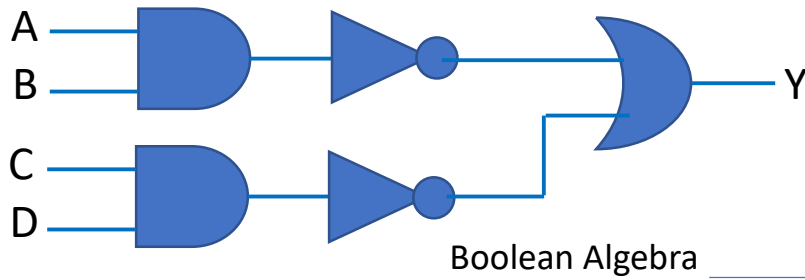
Boolean Algebra _____



Boolean Algebra _____



Logic Gates to algebraic equation *continued...*



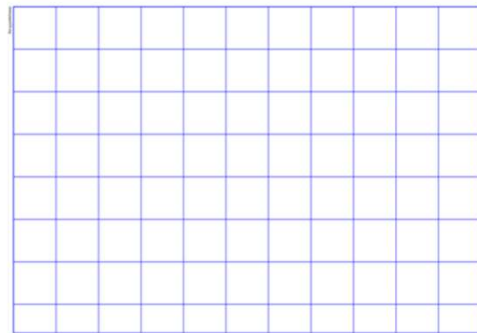
Logic Gates to algebraic equation, with Truth table.
continued...



Boolean Algebra



Boolean Algebra



Boolean Algebra



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